The Core Concept for the Multidimensional Knapsack Problem*

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Abstract. We present the newly developed core concept for the Multidimensional Knapsack Problem (MKP) which is an extension of the classical concept for the one-dimensional case. The core for the multidimensional problem is defined in dependence of a chosen efficiency function of the items, since no single obvious efficiency measure is available for MKP. An empirical study on the cores of widely-used benchmark instances is presented, as well as experiments with different approximate core sizes. Furthermore we describe a memetic algorithm and a relaxation guided variable neighborhood search for the MKP, which are applied to the original and to the core problems. The experimental results show that given a fixed run-time, the different metaheuristics as well as a general purpose integer linear programming solver yield better solution when applied to approximate core problems of fixed size.

1 Introduction

The Multidimensional Knapsack Problem (MKP) is a well-studied, strongly NP-hard combinatorial optimization problem occurring in many different applications. It can be defined by the following ILP:

$$\text{(MKP)} \qquad \text{maximize} \quad z = \sum_{j=1}^{n} p_j x_j \tag{1}$$

$$\text{subject to} \quad \sum_{j=1}^{n} w_{ij} x_j \le c_i, \quad i = 1, \dots, m \tag{2}$$

$$x_j \in \{0, 1\}, \quad j = 1, \dots, n. \tag{3}$$

Given are n items with profits $p_j > 0$ and m resources with capacities $c_i > 0$. Each item j consumes an amount $w_{ij} \ge 0$ from each resource i. The goal is to

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select a subset of items with maximum total profit, see (1); chosen items must, however, not exceed resource capacities, see (2). The 0–1 decision variables x_j indicate which items are selected.

A comprehensive overview on practical and theoretical results for the MKP can be found in the monograph on knapsack problems by Kellerer et al. [8]. Besides exact techniques for solving small to moderately sized instances, see [8], many kinds of metaheuristics have already been applied to the MKP. To our knowledge, the method currently yielding the best results, at least for commonly used benchmark instances, was described by Vasquez and Hao [18] and has recently been refined by Vasquez and Vimont [19]. Various other metaheuristics have been described for the MKP [5, 3], including several variants of hybrid evolutionary algorithms (EAs); see [16] for a survey and comparison of EAs for the MKP.

We first introduce the core concept for KP, and then expand it to MKP with respect to different efficiency measures. We then give some results of an empirical study of the cores of widely-used benchmark instances and present the application of the general ILP-solver CPLEX to MKP cores of fixed sizes. Furthermore we present a Memetic Algorithm (MA) and a Relaxation Guided Variable Neighborhood Search (RGVNS) applied to cores of hard to solve benchmark instances. We finally conclude by summarizing our work.

2 The Core Concept

The core concept was first presented for the classical 0/1-knapsack problem [1], which led to very successful KP algorithms [9, 11, 12]. The main idea is to reduce the original problem by only considering a core of items for which it is hard to decide if they will occur in an optimal solution or not, whereas the variables for all items outside the core are fixed to certain values.

2.1 The Core Concept for KP

The one-dimensional 0/1-knapsack problem (KP) considers items $j = 1, \dots, n$, associated profits p_j, and weights w_j. A subset of these items has to be selected and packed into a knapsack having a capacity c. The total profit of the items in the knapsack has to be maximized, while the total weight is not allowed to exceed c. Obviously, KP is the special case of MKP with $m = 1$.

If the items are sorted according to decreasing efficiency values

$$e_j = \frac{p_j}{w_j}, \tag{4}$$

it is well known that the solution of the LP-relaxation consists in general of three consecutive parts: The first part contains variables set to 1, the second part consists of at most one split item s, whose corresponding LP-values is fractional, and finally the remaining variables, which are always set to zero, form the third part. For most instances of KP (except those with a very special